Josh Steckel

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EDUCATION

University of Delaware *Expected 2029*
Bachelor of Engineering (B.E.) – Electrical Engineering | GPA: 3.99 | Dean's List
Relevant Coursework: FPGA & Digital Design, Digital Signal Processing, Embedded Systems, Circuit Analysis

SKILLS

Core Skills: PCB Design, RF and Digital Signal Processing, FPGA and Digital Design, Embedded Systems, Systems Programming
Hardware & Test Equipment: Oscilloscope, Spectrum Analyzer, Vector Network Analyzer (VNA), Multimeter
Programming Languages: C, C++, Python, MATLAB, Verilog, VHDL, JavaScript
Software & Tools: KiCad, Vivado, GNU Radio, Git, Linux, Visual Studio Code, SolidWorks

EXPERIENCE

Ground Station Team Lead — Delaware Atmospheric Plasma Probe Experiment (DAPPEr) CubeSat *05/2026 – Present*

- Led a 5-member subteam designing and upgrading ground station support equipment for a student-built LEO CubeSat mission measuring ionospheric electron density and temperature over a UHF communications link
- Developed and tested channel models in MATLAB to determine optimal constant and adaptive downlink data rates for the UHF spacecraft communications link, balancing link margin, throughput, and reliability
- Evaluated mission link budget and communications system performance for NASA's CubeSat Launch Initiative Pre-Procurement Readiness Review (PPRR) and FCC licensing, supporting technical documentation and regulatory compliance reviews

Power Side-Channel Attacks Research Assistant — UD Dept. of Electrical & Computer Engineering (VIP) *02/2026 – 07/2026*

- Trained and tested CNN models on 60+ hours of activity data to classify VR device activity from STFT frequency-spectrum images, achieving 97% classification accuracy and strengthening applied signal processing and data analysis skills

Crew Member — AMC Theatres, Inc., Allentown, PA *05/2024 – 08/2024*

- Delivered guest services in a high-volume, fast-paced environment, operating concessions and maintaining a clean, organized, and safety-conscious space while collaborating with a large team

PROJECTS

Altoids Tin Meshtastic Node

- Designed and assembled a 2-layer impedance-controlled RF PCB integrating an ESP32 MCU, UHF LoRa module, GPS module, user interface, and TFT LCD display
- Built and tested a LiPo battery-powered Meshtastic LoRa node housed in an Altoids tin, enabling decentralized, off-the-grid text communications and validating RF performance with lab test equipment

University Transit System Departure Board

- Designed a compact OLED arrival/departure board compatible with any ETA Transit-powered system, built on an automated headless web scraper (Selenium, Docker) hosted on AWS Lambda
- Implemented multi-core RTOS firmware on a low-power ESP32 client fetching pre-processed JSON data over secure HTTPS, cutting load time by 75% while running a horizontally scrolling marquee display

Generative AI Calculator Module

- Developed an embedded AI module enabling natural language math prompting on a TI-84 graphing calculator, programming C/C++ applications for USB serial communication between calculator and module

ACTIVITIES, LEADERSHIP & CERTIFICATIONS

- IEEE — Member
- Amateur Radio Club — Treasurer
- FCC Amateur Radio License – Technician Class (KD3DKR)